AIM6 (Q07716) — Phosphoric Diester Hydrolase Activity (GO:0008081): Hypothesis Deep Research

Gene: AIM6 / YDL237W · *Saccharomyces cerevisiae* (NCBITaxon:559292) **UniProt:** Q07716 (AIM6_YEAST, 390 aa) **Focus:** function_assignment — does AIM6 directly have phosphoric diester hydrolase activity (GO:0008081)? **Evidence type under review:** IEA · **Original reference:** GO_REF:0000002 (InterPro2GO)

Summary

The hypothesis under review is that AIM6 directly possesses **phosphoric diester hydrolase activity (GO:0008081)**. The verdict is **partially supported as a computational prediction, but not experimentally established**. AIM6 genuinely belongs to the phospholipase-C-like (PI-PLC/GDPD) TIM β/α-barrel superfamily, and a structural analysis of its AlphaFold model performed in this review shows the catalytic scaffold appears intact — a His118–His120 dyad at 5.5 Å (canonical PLC/GDPD catalytic-histidine geometry) ringed by Glu141, Arg350 and several aspartates, all modeled with high local confidence (pLDDT ≈ 89). On fold and geometry grounds, AIM6 is therefore **unlikely to be an obvious pseudo-enzyme**, and GO:0008081 is a defensible predicted molecular-function hypothesis.

However, three independent lines of evidence keep the annotation firmly in "predicted/unverified" territory. First, the GO:0008081 IEA is inherited from the **broad superfamily entry (IPR017946)**, not from the AIM6-specific family; the AIM6-specific InterPro domain (IPR039559), the AIM6-specific Pfam family (PF30387), and the matching CDD model (cd08577) all deliberately assert **no molecular function**. Second, AIM6 itself is an **uncharacterized protein of unknown function** — SGD and UniProt document only a respiratory-growth defect on disruption, with no assay, substrate, product, or EC number. Third, a **pan-family audit found that no AIM6 ortholog anywhere carries a characterized catalytic activity**: 0 of 7 reviewed and 0 of ~200 unreviewed members have an EC number or catalytic-activity

comment, and all carry the identical single GO:0008081 term via the same automated pipeline — the classic signature of uniform database carry-over rather than convergent experimental support.

Bottom line for the curator: Retain GO:0008081 as a **low-confidence, predicted IEA**, flagged as unverified. Do not upgrade it to experimentally supported, and do not substitute a narrower term (e.g., PI-PLC or GDPD activity) without a direct enzymatic assay. The documented biology — respiratory growth and mitochondrial inheritance — is a downstream loss-of-function phenotype that has never been mechanistically linked to phosphodiesterase chemistry.

Key Findings

Finding 1 — GO:0008081 is a fold-based IEA from the broad PLC-like superfamily, not the AIM6-specific family

QuickGO records that Q07716 carries **GO:0008081 (phosphoric diester hydrolase activity)** with evidence code **IEA/ECO:0000256**, reference **GO_REF:0000002 (InterPro2GO)**, and `withFrom` **InterPro:IPR017946**. IPR017946 is *"PLC-like phosphodiesterase, TIM beta/alpha-barrel domain superfamily"* — a **homologous_superfamily** entry, the broadest and least specific tier of InterPro signature. Its InterPro2GO mapping projects two terms: the molecular-function term GO:0008081 and the biological-process term GO:0006629 (lipid metabolic process).

Critically, the **AIM6-specific signatures carry no GO terms at all.** The dedicated family entry **IPR039559** (*"Altered inheritance of mitochondria protein 6, PI-PLC-like catalytic domain"*) and the dedicated Pfam domain **PF30387** (*"AIM6-like phosphodiesterase domain"*) have **zero** associated GO annotations. The protein also matches **CDD cd08577** — explicitly described as *"Uncharacterized hypothetical proteins similar to the catalytic domains of PI-PLC and glycerophosphodiester phosphodiesterases"* — and superfamily **SSF51695**. In other words, the curators who built the AIM6-specific models deliberately declined to assert a function; GO:0008081 appears only because AIM6 also trips the umbrella superfamily filter. This is a textbook fold-based over-projection scenario, where the specific-family evidence is weaker than a typical InterPro2GO IEA.

Finding 2 — AIM6 is an uncharacterized protein required for respiratory growth, with no direct assay of phosphodiesterase activity

SGD describes YDL237W/AIM6 as a *"Protein of unknown function; required for respiratory growth; YDL237W is not an essential gene."* UniProt Q07716 (AIM6_YEAST, a 390-aa precursor bearing a predicted N-terminal signal/targeting sequence) contains **no FUNCTION comment and no CATALYTIC ACTIVITY comment**. Its only substantive functional annotations are a **DISRUPTION PHENOTYPE** ("Impairs respiratory growth," from PMID: 19300474, the systematic *Altered Inheritance of Mitochondria* screen that named the gene) and a **SIMILARITY** statement ("Belongs to the AIM6 family").

The InterPro description for the AIM6 family (IPR039559) is unusually candid: *"the PI-PLC-like catalytic domain found in baker's yeast Aim6... The function of Aim6 is not clear."* There are **no experimental 3D structures** for AIM6 (InterPro structures = 0). The only structural information available is the **AlphaFold model AF-Q07716-F1**, of good overall quality (mean pLDDT = 88) but a prediction, not empirical data. Thus the single experimentally anchored fact about AIM6 — a respiratory-growth defect — is a **downstream loss-of-function phenotype** never connected to phosphodiesterase chemistry, and is equally consistent with a structural, regulatory, or non-catalytic scaffolding role.

Finding 3 — No AIM6-family member has any characterized catalytic activity; GO:0008081 is uniform database carry-over

A pan-family audit queried UniProt for the entire AIM6 family (InterPro IPR039559). **Zero of 7 reviewed** members and **zero of 200 sampled unreviewed** members carry an EC number or a CATALYTIC ACTIVITY comment. All seven reviewed members — Q07716, A6ZX97, B3LHB3, C7GJQ6, C8Z6N5, B5VF44 (five *S. cerevisiae* strain entries) plus Q75E59 (*Eremothecium gossypii*) — are named *"Altered inheritance of mitochondria protein 6,"* have **no FUNCTION comment**, and carry the **identical single MF annotation GO:0008081** via the same IEA pipeline. A parallel query of Pfam PF30387 returned **0 reviewed proteins with an EC number or catalytic-activity comment**.

This uniformity is diagnostically important. When a real enzymatic activity has been established for a protein family, one expects at least one member to carry experimental (IDA/IMP) evidence, an EC number, defined substrates, or kinetic data that then propagate as a curated inference. Here every member carries exactly one term, from exactly one automated source, with no experimental anchor anywhere in the family — the signature of **fold-based propagation across an entirely uncharacterized family**, not convergent independent support.

Finding 4 — The AlphaFold model retains a PLC/GDPD-like active-site residue cluster

Structural analysis of AF-Q07716-F1 (v6; 3,131 atoms parsed) shows the model is not an obviously degraded pseudo-enzyme. Among its 10 histidines, the closest imidazole pair, **His118–His120**, sits at **5.49 Å N–N** — consistent with the **catalytic histidine dyad of the PI-PLC/GDPD-like superfamily** (bacterial PI-PLC catalytic His pair ~5–6 Å apart). The dyad centroid is surrounded (within 9 Å) by a constellation typical of this catalytic architecture: **Glu141 (3.1 Å), Arg350 (2.9 Å), Asp122 (7.7 Å), Asp143 (7.1 Å), Asp210 (7.9 Å), and Asp381 (5.8 Å)**. The local sequence context is an **HxH motif** (His118-Ser119-His120) followed by Asn-Asp (…VHSHNDYW…), and the **local pLDDT at the dyad ≈ 89** (high confidence).

This is the strongest single piece of *positive* evidence for the hypothesis: residues required for PLC-type phosphodiester hydrolysis appear spatially organized in the predicted structure, elevating AIM6 above a bare fold match. Nevertheless it remains **inference from a predicted model** — residue proximity is necessary but not sufficient for catalysis; substrate specificity, metal/cofactor requirements, and the true reaction cannot be read from geometry alone.

Mechanistic Model / Interpretation

The evidence assembles into a layered picture in which support decays as one moves from "belongs to a catalytic superfamily" toward "is itself a phosphodiesterase acting on a specific substrate in vivo."

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LAYER OF EVIDENCE                          STRENGTH    WHAT IT ACTUALLY SHOWS
------------------------------------------ ----------- ------------------------------------------
Superfamily fold (IPR017946, SSF51695)     Strong      AIM6 has a PLC-like TIM β/α-barrel fold
AlphaFold active-site scaffold intact       Moderate    His118–His120 dyad + acidic/Arg cluster
  (His dyad 5.5 Å, pLDDT ~89)                           present in the *predicted* model
------------------------------------------ ----------- ------------------------------------------
AIM6-specific models (IPR039559, PF30387,   ABSENT      Curators assign NO function;
  CDD cd08577) assign a function                        "The function of Aim6 is not clear."
Any family member with EC / activity /      ABSENT      0/7 reviewed, 0/200 unreviewed
  substrate / kinetics
Direct enzymatic assay of AIM6              ABSENT      No substrate, product, or rate measured
Link from phosphodiesterase chemistry to    ABSENT      Respiratory-growth defect unexplained
  the respiratory-growth phenotype
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Interpretation. AIM6 is best modeled as a **member of the PLC/GDPD-like superfamily whose specific catalytic function has never been demonstrated and may differ from the ancestral activity.** GO:0008081 is a legitimate *hypothesis-generating* prediction — the fold is real and the active-site scaffold is preserved — but it currently rides entirely on homology, and the deliberate silence of every AIM6-specific curated model is a strong signal that expert curators regard the activity as unestablished.

Two mechanistic scenarios remain open and are not distinguished by present data. (1) **Active enzyme:** AIM6 hydrolyzes a phosphodiester bond (in a lipid, a glycerophosphodiester, or another substrate) as part of a mitochondrial/respiratory function. (2) **Non-catalytic or neofunctionalized role:** AIM6 retains the fold and even active-site-like residues but functions as a lipid-binding/scaffolding or regulatory protein in mitochondrial inheritance, with the respiratory phenotype arising from that role. The documented biology (mitochondrial inheritance, respiratory growth; PMID: 19300474) is compatible with either. A localization mismatch adds caution: AIM6 is a precursor with an N-terminal signal/anchor and a mitochondrial/respiratory phenotype, whereas classical PI-PLC signaling is cytoplasmic/nuclear.

Importantly, the well-characterized yeast PLC pathway is carried out by **Plc1p**, not AIM6. The reviewed literature (PMIDs 23381992, 23179856, 19459978, 40172212) establishes that phosphoinositide-specific PLC signaling in *S. cerevisiae* is a Plc1-driven, nucleus-associated pathway feeding inositol-polyphosphate synthesis; none implicates AIM6. AIM6 is therefore **not** the recognized cellular phosphodiesterase, and its predicted activity, if real, is a distinct and uncharacterized one.

Evidence Base (Evidence Matrix)

#	Citation / Source	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Cont
1	QuickGO record for Q07716 (GO_REF:0000002)	Database/ computational	Qualifies	On what basis is GO:0008081 assigned?	GO:0008081 is IEA (ECO:0000256), withFrom **InterPro:IPR017946** (superfamily), plus IEA GO:0006629	*S. cer* prote
2	InterPro IPR017946 (superfamily) InterPro2GO	Structural/ evolutionary (homology)	Supports (fold-level)	Does the fold map to the activity?	*PLC-like phosphodiesterase TIM β/α-barrel superfamily* → GO:0008081 + GO:0006629	Cross king supe incl. PLCs
3	InterPro IPR039559 + Pfam PF30387	Structural/ evolutionary	**Qualifies / cautionary**	Does the *AIM6-specific* family assert the activity?	AIM6-specific domain "PI-PLC-like catalytic domain"; **"function of Aim6 is not clear"**; **no GO term**; 0 experimental structures	Fung euka AIM
4	CDD cd08577	Structural/ evolutionary	Qualifies (cautionary)	How confident is the domain model?	Title: **"Uncharacterized hypothetical proteins similar to the catalytic domains of PI-PLC and glycerophosphodiester phosphodiesterases"**	Cons dom data
5	UniProt Q07716 (AIM6_YEAST)	Curated record / mutant phenotype	Qualifies	What is experimentally known?	Only DISRUPTION PHENOTYPE "Impairs respiratory growth" + SIMILARITY; **no CATALYTIC ACTIVITY/ FUNCTION** comment; 390 aa precursor	*S. cer*
6	SGD YDL237W	Curated database	Qualifies	Does the MOD assert the activity?	**"Protein of unknown function; required for**	*S. cer*

#	Citation / Source	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Cont
					respiratory growth", non-essential	
7	AlphaFold AF-Q07716-F1	Structural (predicted)	Supports (fold present)	Is the PLC-like TIM barrel present?	Model mean pLDDT = 88; no experimental structure exists	Com
8	PMID: 19300474	Mutant phenotype	Qualifies	Source of the phenotype annotation	AIM6 deletion alters mitochondrial inheritance / impairs respiratory growth	*S. cer* geno scree
9	PMID: 23381992, 23179856, 19459978	Primary literature	Competing (paralog/ context)	Which yeast protein is the known PI-PLC?	Yeast phosphoinositide PLC activity is carried by **Plc1**, a separate gene	*S. cer*
10	UniProt/InterPro family audit (IPR039559 / PF30387)	Computational (database audit)	**Qualifies (over-annotation signal)**	Does any family member have a characterized activity?	**0/7 reviewed and 0/200 unreviewed** members carry EC or catalytic-activity comment; identical GO:0008081 IEA across orthologs incl. *E. gossypii* Q75E59	Pan- (Fun
11	AlphaFold AF-Q07716-F1 active-site analysis (this run)	Structural (predicted)	**Supports (weakly, geometry-level)**	Is the catalytic-residue scaffold retained?	His118–His120 dyad at 5.49 Å; Glu141 (3.1 Å), Arg350 (2.9 Å), Asp122/143/210/381; HxH-ND motif; local pLDDT ≈ 89	Com (sing

Literature notes. The literature consistently identifies **Plc1p** — not AIM6 — as the experimentally validated phosphoinositide phospholipase C of *S. cerevisiae* (PMID: 23179856, PMID: 19459978), with the canonical $PIP_2 \rightarrow IP_3$ chemistry described comparatively in PMID: 23381992. PMID: 40172212 shows PLC-independent inositol-polyphosphate routes also exist, broadening context. None of these papers implicates AIM6, reinforcing that the seed hypothesis rests on homology/structure, not any direct study of AIM6.

GO Curation Implications

Lead (requires curator verification): Retain GO:0008081 as an IEA, flagged low-confidence / predicted-unverified. Do not upgrade; do not narrow without a direct assay.

- **Aspect — Molecular Function (GO:0008081):** biologically *plausible from fold homology* but **not experimentally supported**. Because the assignment originates from the **superfamily** entry (IPR017946) while the **AIM6-specific** family (IPR039559/PF30387) carries no GO mapping, evidence for the *specific* activity is weaker than a typical InterPro2GO IEA. A curator could justifiably keep it as a low-confidence prediction or mark it as potential over-annotation pending catalytic-residue verification.
- **Companion term — GO:0006629 (lipid metabolic process, BP):** same InterPro2GO mapping, equally fold-derived and unverified; treat identically (non-core).
- **Do NOT make more specific:** GO:0004435 (phosphatidylinositol phospholipase C) or GO:0008889 (glycerophosphodiester phosphodiesterase) are the two catalytic identities the fold hints at, but neither is supported by data. Do not substitute a more specific term.
- **Do NOT default to "protein binding":** the informative, honest state here is "predicted PLC-like phosphodiesterase fold, activity unverified" — GO:0008081 (unproven) is more informative and consistent with the fold.

GO decision table

Term	Aspect	Current	Recommended action	Rationale
GO:0008081 phosphoric diester hydrolase activity	MF	IEA (GO_REF:0000002, withFrom IPR017946)	**Retain, mark predicted/low-confidence**	Real fold + intact predicted active site, but no assay; family-wide carry-over
GO:0006629 lipid metabolic process	BP	IEA (same mapping)	**Retain as unverified / non-core**	Same weak provenance; no experimental link
GO:0004435 / GO:0008889 (specific children)	MF	not assigned	**Do not assert**	No substrate/assay data to select a child term
respiratory growth / mito inheritance	BP	phenotype only	**Do not annotate as direct function**	Downstream loss-of-function phenotype, not activity

Mechanistic Scope

The immediate molecular function under test is **hydrolysis of a phosphoric diester bond** (EC 3.1.4.-; GO:0008081) — the chemistry shared by phospholipase C and glycerophosphodiester phosphodiesterase enzymes of the PLC/GDPD-like TIM-barrel superfamily. The direct gene-product-level question is: *does AIM6 itself cleave a phosphodiester substrate?*

What is **direct** in the evidence base: the fold assignment (superfamily match) and the predicted spatial organization of candidate catalytic residues (His118–His120 dyad plus an acidic/Arg constellation). What is **downstream or inferred**: the respiratory-growth defect on gene disruption (PMID: 19300474) is a whole-cell loss-of-function phenotype that could arise from any essential-for-respiration role, catalytic or not; the "lipid metabolic process" BP term is a computational inference from the same fold, not an observed metabolic change; and the mitochondrial-inheritance context is a genetic-screen association, not a demonstrated activity. No substrate, product, rate, cofactor, or EC number has ever been measured for AIM6 or any family member — so the mapping from "has the fold" to "performs the reaction in vivo" remains entirely unbridged by experiment.

Conflicts and Alternatives

1. **Superfamily-vs-family discordance (strongest caveat):** the activity is asserted only at the superfamily level; the AIM6-specific family declines it. Divergent superfamily members frequently retain the fold but lose catalysis or acquire new functions.
2. **Pseudo-enzyme / neofunctionalization alternative:** AIM6 could retain the fold (and even active-site-like residues) while acting as a non-catalytic lipid-binding scaffold or regulator in mitochondrial inheritance. The intact AlphaFold scaffold weakens but does not eliminate this — many pseudo-enzymes retain intact-looking active sites.
3. **Paralog/context confusion with Plc1p:** yeast's characterized phosphoinositide-PLC activity is carried by **Plc1** (PMID: 23179856, PMID: 19459978); the "yeast has a PLC" intuition should not be transferred to AIM6.
4. **Localization mismatch:** AIM6 is a precursor with an N-terminal signal/anchor and a mitochondrial/respiratory phenotype, whereas classical PI-PLC signaling is cytoplasmic/nuclear.
5. **Database carry-over risk (now demonstrated):** GO_REF:0000002 pipelines propagate superfamily activities broadly. The pan-family audit found **0/7 reviewed and 0/~200 unreviewed** members with any EC or catalytic-activity annotation — the identical GO:0008081 IEA stamped uniformly across orthologs. This is the signature of computational carry-over, confirmed by the "uncharacterized" CDD labelling.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Catalytic-residue conservation	AlphaFold active-site analysis found a retained His118–His120 dyad (5.5 Å) with Glu141/Arg350/Asp cluster (local pLDDT ≈ 89). A formal structural superposition onto a reference PI-PLC/GDPD was not run (no alignment tool available)	Scaffold looks intact (argues against pseudo-enzyme), but geometry alone does not prove catalysis or identify a substrate	Formal superposition (phenix.superpose / DALI) of AF-Q07716-F1 onto bacterial PI-PLC / GDPD; residue-by-residue equivalence; metal site check
Actual substrate	No enzymatic assay found in literature	PI-PLC vs GDPD vs other substrate determines the correct specific GO term	In vitro assay against PI/PIP_2, glycerophosphodiesters, generic phosphodiesters
Subcellular location	UniProt notes signal/precursor; phenotype is respiratory	Determines whether a phosphodiesterase role is plausible in situ	GFP localization / fractionation / proteomics
Direct vs indirect respiratory role	Only deletion phenotype known	BP context (lipid metabolism vs mitochondrial biogenesis) hinges on this	Metabolomics/lipidomics of *aim6Δ*; suppressor screen
Post-annotation experimental data	UniProt/SGD/InterPro queried	New data could break the uniform no-data pattern	Literature/database re-check at review time

Additional limitations of this review: no wet-lab data on AIM6 enzymatic activity exists in the accessed literature; conclusions rest on sequence/fold databases, an AlphaFold model, and one phenotype paper. The active-site argument derives from a predicted structure, not an experimental one. All computational lookups (UniProt, InterPro, QuickGO, AlphaFold, SGD) were reported as retrieved, not fabricated.

Discriminating Tests

1. **Catalytic-site structural alignment (fastest):** superpose AF-Q07716-F1 onto a bacterial PI-PLC (e.g., *B. cereus*) and a GDPD; test whether the conserved catalytic histidine pair and acidic/metal-binding residues are spatially conserved. Absence → refute (pseudo-enzyme).
2. **Active-site mutagenesis** + **functional rescue:** replace His118 and/or His120 with Ala and test whether the mutant still complements the respiratory-growth defect of *aim6Δ*. Loss of rescue links catalysis to physiology; retention argues for a non-catalytic role.
3. **Direct biochemistry:** recombinant AIM6 assayed for hydrolysis of PI/PIP_2 (PLC readout) and glycerophosphodiesters (GDPD readout), with metal/cofactor titration.
4. **Comparative metabolomics:** *aim6Δ* vs wild type, focused on phosphoinositides and glycerophosphodiesters; an accumulating substrate would nominate the true reaction.
5. **Cross-family check:** assay a tractable ortholog (e.g., *E. gossypii* Q75E59) to test whether any family member shows activity, breaking the uniform no-data pattern.

Curation Leads (require curator verification)

- **Reference to verify:** InterPro **IPR039559** description — exact snippet: *"the PI-PLC-like catalytic domain found in baker's yeast Aim6 ... The function of Aim6 is not clear."* Confirms family-level curators withhold a specific function.
- **Reference to verify:** CDD **cd08577** title — *"Uncharacterized hypothetical proteins similar to the catalytic domains of Phosphoinositide-specific phospholipase C and Glycerophosphodiester phosphodiesterases."*
- **Reference to verify:** SGD YDL237W — *"Protein of unknown function; required for respiratory growth."*

- **Primary reference for phenotype (not function):** PMID: 19300474 — source of "impairs respiratory growth."
- **Provenance to note:** QuickGO shows GO:0008081 on Q07716 with `evidence = IEA/ECO:0000256`, `reference = GO_REF:0000002`, `withFrom = InterPro:IPR017946`; the AIM6-specific IPR039559, Pfam PF30387, and CDD cd08577 carry **no** GO terms.
- **Candidate action change:** keep GO:0008081 as **IEA/predicted-uncertain**; flag as possible over-annotation; do **not** add experimental evidence; do **not** substitute a narrower term (GO:0004435 / GO:0008889) without assay data. Apply the same treatment to the co-inherited IEA GO:0006629.
- **Suggested curator question:** "Are AIM6's PLC/GDPD catalytic residues conserved across the family, and does a catalytic-dead allele fail to rescue *aim6Δ*?" — the single most decisive check.
- **Suggested experiment:** recombinant phosphodiesterase assay (PI/PIP_2 and glycerophosphodiester substrates) + His118/His120→Ala rescue of *aim6Δ* + *aim6Δ* mitochondrial lipidomics.

Proposed Follow-up Actions

1. **Curator:** Retain GO:0008081 (and GO:0006629) as predicted/low-confidence IEA; annotate provenance (IPR017946 superfamily tier; family-specific models carry no function). Do not upgrade or narrow.
2. **Wet-lab flag:** Prioritize the His118/His120→Ala rescue experiment and an in vitro phosphodiesterase assay — the two tests that would most directly settle the hypothesis.
3. **Bioinformatics flag:** If a repository `*-bioinformatics` analysis exists, compare its active-site and family-audit conclusions against Findings 3–4 here.
4. **Re-review trigger:** Revisit if any AIM6-family member acquires an experimental catalytic-activity annotation or EC number, which would break the current uniform-carry-over pattern.

Verdict: Partially supported (computational prediction only). AIM6 has a real PLC/GDPD-like fold and an intact predicted active site, but the GO:0008081 annotation is fold-based superfamily carry-over with no experimental support in AIM6 or any family member. Retain as low-confidence IEA pending a direct assay.
